

Christopher Musco

370 Jay Street, Floor 11 – Brooklyn, NY 11201 – Office 1105

☎ (401) 578 2541 • ✉ cmusco@nyu.edu • 🌐 www.chrismusco.com

Academic Positions

New York University <i>Assistant Professor, Tandon School of Engineering</i> <i>Department of Computer Science and Engineering</i>	New York, NY <i>2019 - present</i>
Microsoft Research <i>Visiting Researcher, Machine Learning and Optimization Group</i>	Redmond, WA <i>Summer 2020</i>
Princeton University <i>Research Instructor, Department of Computer Science</i>	Princeton, NJ <i>2018 – 2019</i>
Massachusetts Institute of Technology <i>Research and Teaching Assistant, Theory of Computation Group</i>	Cambridge, MA <i>2013 – 2018</i>
Yahoo Labs <i>Research Intern, Scalable Machine Learning Group</i>	New York, NY <i>Summer 2015</i>

Research Areas

theory of algorithms • randomization • numerical linear algebra • sketching and streaming • foundations of data science • network science • active machine learning • spectral graph algorithms •

Education

Massachusetts Institute of Technology <i>Ph.D. in Computer Science</i> Advisor: Jonathan Kelner Ph.D. Thesis: Faster Linear Algebra for Data Analysis and Machine Learning M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices	Cambridge, MA <i>2013 – 2018</i>
Yale University <i>B.S. in Applied Mathematics, B.S. in Computer Science</i>	New Haven, CT <i>2008 – 2012</i>

Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a *, which indicates that authors are ordered by contribution.*

Near-Optimality Guarantees for Approximating Rational Matrix Functions by the Lanczos Method. Noah Amsel, Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *Preprint* 2023.

Weighted Minwise Hashing Beats Linear Sketching for Inner Product Estimation. Aline Bessa, Majid Daliri, Juliana Freire, Cameron Musco, Christopher Musco, Aécio Santos, Haoxiang Zhang. *ACM Symposium on Principles of Database Systems (PODS)* 2023.

Low-memory Krylov subspace methods for optimal rational matrix function approximation. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *SIAM Journal on Matrix Analysis and Applications*

(SIMAX) 2023.

Active Learning for Single Neuron Models with Lipschitz Non-Linearities. Aarshvi Gajjar, Chinmay Hegde, Christopher Musco. *International Conference on Artificial Intelligence and Statistics (AISTATS)* 2023. Preliminary version at NeurIPS 2022 Workshop on The Symbiosis of Deep Learning and Differential Equations II (DLDE 2022).

Near-Linear Sample Complexity for L_p Polynomial Regression. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff, Samson Zhou. *ACM-SIAM Symposium on Discrete Algorithms (SODA)*. 2023.

A Tight Analysis of Hutchinson's Diagonal Estimator. Prathamesh Dharangutte, Christopher Musco. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2023.

Active Linear Regression for ℓ_p Norms and Beyond. Cameron Musco, Christopher Musco, David P. Woodruff, Taisuke Yasuda. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2022.

How to Quantify Polarization in Models of Opinion Dynamics. Christopher Musco, Indu Ramesh, Johan Ugander, R. Teal Witter. *International Workshop on Mining and Learning with Graphs (at KDD)* 2022.

Chemically-informed data-driven optimization (ChIDDO): Leveraging physical models and Bayesian learning to accelerate chemical research*. Daniel Frey, Juhee Shin, Christopher Musco, Miguel A. Modestino. *Reaction Chemistry & Engineering* 2022.

Streaming Approach to In Situ Selection of Key Time Steps for Time-Varying Volume Data*. Mengxi Wu, Yi-Jen Chiang, Christopher Musco. *EuroVis* 2022.

Sublinear Time Spectral Density Estimation. Vladimir Braverman, Aditya Krishnan, Christopher Musco. *ACM Symposium on Theory of Computing (STOC)* 2022.

A Sketch-based Index for Correlated Dataset Search*. Aécio Santos, Aline Bessa, Christopher Musco, Juliana Freire. *IEEE 38th International Conference on Data Engineering (ICDE)* 2022.

Fast Regression for Structured Inputs. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff, Samson Zhou. *International Conference on Learning Representations (ICLR)* 2022.

Error Bounds for Lanczos-based Matrix Function Approximation. Tyler Chen, Anne Greenbaum, Cameron Musco, Christopher Musco. *SIAM Journal on Matrix Analysis and Applications (SIMAX)* 2022.

Fast and Near-Optimal Diagonal Preconditioning. Arun Jambulapati, Jerry Li, Christopher Musco, Aaron Sidford, Kevin Tian. *Preprint* 2021.

Dynamic Trace Estimation. Prathamesh Dharangutte, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2021.

Finding an Approximate Mode of a Kernel Density Estimate. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *European Symposium on Algorithms (ESA)* 2021.

Correlation Sketches for Approximate Join-Correlation Queries*. Aécio Santos, Aline Bessa, Fernando Chirigati, Christopher Musco, Juliana Freire. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Public Transport Planning: When Transit Network Connectivity Meets Commuting Demand*. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *ACM Special Interest Group on Management of Data Conference (SIGMOD)* 2021.

Simple Heuristics Yield Provable Algorithms for Masked Low Rank Approximation. Cameron Musco, Christopher Musco, David Woodruff. *Innovations in Theoretical Computer Science (ITCS)* 2021.

Hutch++: Optimal Stochastic Trace Estimation. Raphael Meyer, Cameron Musco, Christopher Musco, David P. Woodruff. *SIAM Symposium on Simplicity in Algorithms (SOSA)* 2021.

Graph Learning for Inverse Landscape Genetics. Prathamesh Dharangutte, Christopher Musco. *AAAI Conference on Artificial Intelligence (AAAI)* 2021. Short version in NeurIPS 2020 AI for Earth Sciences workshop.

Fourier Sparse Leverage Scores and Approximate Kernel Learning. Tamás Erdélyi, Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020. Invited for spotlight presentation (4% of submissions).

Near Optimal Linear Algebra in the Online and Sliding Window Models. Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2020.

The Statistical Cost of Robust Kernel Hyperparameter Tuning. Raphael Meyer, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.

Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

Sample Efficient Toeplitz Covariance Estimation. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Fast and Space Efficient Spectral Sparsification in Dynamic Streams. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Analyzing the Impact of Filter Bubbles on Social Network Polarization. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM)* 2020. Preliminary version in KDD Workshop on Issues of Sentiment Discovery and Opinion Mining.

Projection-Cost-Preserving Sketches: Proof Strategies and Constructions . Cameron Musco, Christopher Musco. *Technical Report* 2020.

A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

Inferring networks from random walk-based node similarities. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

Eigenvector Computation and Community Detection in Asynchronous Gossip Models. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

Minimizing Polarization and Disagreement in Social Networks. Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)*

2015. Invited for full oral presentation (1 of 15, < 1% of submissions). Also presented at the 2016 Copper Mountain Conference on Iterative Methods.

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)*. 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Funding

NSF CAREER Award

Frontiers in Matrix Sketching.

2021-2026

PI, \$562,224

DOE Office of Science Grant

Leverage Score Sampling for Parametric PDEs.

2021-2023

PI (w/ Chinmay Hegde, NYU), \$300,000

NSF III Medium

Dataset Search and Ranking for Data Augmentation and Explanation.

2021-2024

Co-PI (w/ Juliana Freire, NYU), \$1,093,195

Current Graduate Student and Postdoc Advising

Tyler Chen. Postdoc, New York University.

2022 - present

David Persson. Visiting Ph.D. student.

2023

Noah Amsel. Ph.D., New York University (with Prof. Joan Bruna).

2022 - present

Majid Daliri. Ph.D., New York University.

2022 - present

Aarshvi Gajjar. Ph.D., New York University (with Prof. Chinmay Hedge).

2021 - present

Apoorv Singh. Ph.D., New York University.

2020 - present

R. Teal Witter. Ph.D., New York University (with Prof. Lisa Hellerstein).

2020 - present

Raphael Meyer. Ph.D., New York University.

2019 - present

Siddharth Sagar. M.S., New York University.

2023

Yajur Ahuja. M.S., New York University.

2023

Sundeep Chenreddy M.S., New York University.

2023

Atsushu Shimuzu. M.S., New York University.

2022 - present

Danrong Li. M.S., New York University.

2022 - present

Past Advising

Aline Bessa. Postdoc, New York University.

2021

Indu Ramesh. M.S., New York University.	2020 - 2022
Xinyu Luo. M.S., New York University.	2020 - 2022
Prathamesh Dharangutte. M.S., New York University.	2019 - 2021
John Shin. M.S., New York University.	2019 - 2020

I have advised over a dozen undergraduate guided studies and research projects.

Ph.D. Thesis Committees

Francis Williams, NYU	<i>Defended 2021.</i>
Yiming Zhang , NYU	<i>Defending 2022.</i>

Teaching

Machine Learning, Graduate (NYU CS-GY 6923) <i>Course Instructor</i>	<i>Spring 2022, 2023</i>
Algorithmic Machine Learning and Data Science (NYU CS-GY 6763) <i>Course Instructor</i>	<i>Fall 2019, 2020, 2021, 2022</i>
Introduction to Machine Learning (NYU CS-UY 4563) <i>Course Instructor</i>	<i>Spring 2020</i>
Advanced Algorithm Design (Princeton COS 521) <i>Course Instructor</i>	<i>Fall 2018</i>
Advanced Algorithms (MIT 6.854/18.415) <i>Teaching Assistant</i>	<i>Spring 2016</i>
Technical Communication Skills for Graduate Students (MIT 6.S977) <i>Workshop Leader</i>	<i>Spring 2016</i>

Mentorship and Outreach

SIAM/MathWorks Math Modeling Challenge <i>Problem Development Committee, Director of Technical Computing Judging</i>	<i>2018 – present</i>
Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Lead judging and award presentation for student Technical Computing prizes.	
MIT Graduate Communication Lab <i>Communication Adviser</i>	<i>2015 – 2018</i>
Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.	
CSAIL Algorithms Office Hours <i>Member</i>	<i>2016 – 2018</i>
Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.	
Research Science Institute at MIT <i>Summer Research Adviser</i>	<i>2014</i>
Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.	

Invited Talks

Tutorial on Matrix Sketching

BIRS Workshop on Perspectives on Matrix Computations *Mar. 2023*

Integrating Education and Research in Early Career Funding Proposals

Joint Mathematics Meetings, SIAM Special Session on Education *Jan. 2023*

Iterative Sampling Methods for Matrix Approximation

University of Michigan, Randomized Numerical Linear Algebra Seminar *Oct. 2022*

Noisy Implicit Methods for Matrix Functions

SIAM Annual Meeting *July 2022*

Sublinear Time Spectral Density Estimation

Queens College CUNY Computer Science Colloquium *March 2023*

National University of Singapore, Workshop on Algorithms and Foundations for Data Science *June 2022*

Oxford University, Computational Mathematics and Applications Seminar *Feb. 2022*

University of British Columbia, Applied Math Seminar *Jan. 2022*

Workshop on Algorithms for Large Data *Aug. 2021*

Cornell University, Scientific Computing and Numerics Seminar *Mar. 2021*

Optimal Stochastic Trace Estimation

New York University, Numerical Analysis and Scientific Computing Seminar *April 2022*

University of Michigan, Randomized Numerical Linear Algebra Seminar *Nov. 2021*

Stanford University, Operations Research Seminar *Oct. 2020*

A TCS Perspective on Iterative Eigensolvers

Complexity of Matrix Computations Seminar *Oct. 2021*

Mathematical Modeling and Coding in the Classroom

Mu Alpha Theta National Convention *July 2021*

Randomized Algorithms for Probing the Spectrum of a Matrix

George Mason University, AI Tea Series *Apr. 2021*

Importance Sampling for Approximating High-Dimensional Kernel Matrices

Emory University, Numerical Analysis and Scientific Computing Seminar *Feb. 2021*

Structured Covariance Estimation

Oxford Statistics and Machine Learning in Finance Seminar *June 2022*

Yahoo! Research, Seminar *April 2020*

Johns Hopkins University, Theory Seminar *Oct. 2019*

From Randomized Linear Algebra to Randomized Functional Analysis

SIAM Mathematics of Data Science Conference *May 2020*

University of Utah, Data Science Seminar *Jan. 2020*

Rutgers University, DIMACS RandNLA Workshop *Sept. 2019*

Learning Signals with Simple Fourier transforms

Columbia University, Theory Seminar *Oct. 2018*

UC Berkeley, Simons Institute Workshop *Sept. 2018*

Princeton University, Theory Seminar *Sept. 2018*

The Lanczos Method in Data Science: Challenges and Opportunities

SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium	<i>Feb. 2020</i>
Georgetown University, CS Seminar	<i>Feb. 2018</i>
Unifying Random Features and Leverage Scores for Kernel Matrix Approximation	
Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar	<i>Feb. 2018</i>
Harvard University, guest lecture in “Sketching Algorithms for Big Data”	<i>Nov. 2017</i>
Microsoft Research, Redmond, Research Seminar	<i>Sept. 2017</i>
Sketching and Sampling Methods for Large Scale Linear Algebra	
Twitter, Cambridge, Research Seminar	<i>July 2017</i>
Recursive Sampling for the Nyström Method	
New England Machine Learning Day	<i>May 2017</i>
Introduction to Linear Sketching	
MIT, Theory Retreat	<i>Sept. 2016</i>
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	<i>July 2016</i>
MIT, Algorithms & Complexity Seminar	<i>July 2016</i>
University of Utah, Data Group Meeting	<i>Jan. 2016</i>
Large Scale Column Subset Selection	
Yahoo Labs, Science Week	<i>July 2015</i>
Dimensionality Reduction for k-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	<i>Aug. 2015</i>
MIT, Theory Lunch	<i>June 2014</i>
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	<i>June 2015</i>
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day	<i>Apr. 2015</i>
Harvard University, Theory Seminar	<i>Nov. 2014</i>

Service

Organization.....	
NYU Theoretical Computer Science Weekly Seminar	<i>2022 - present</i>
NYU Computer Science Theory “Pandemic Presentations” Workshop	<i>2022</i>
SIAM Annual Meeting Minisymposium Organizer	<i>2021</i>
SIAM Mathematics of Data Science Minisymposium Organizer	<i>2020</i>
Department Service.....	
Faculty Search Committee	<i>2019, 2021, 2022</i>
Department Chair Search Committee	<i>2021 - present</i>
Online Faculty Oversight Committee	<i>2021 - present</i>
Chair, Ph.D. Admissions Committee	<i>2019 - present</i>
Conference Programming.....	
STOC Program Committee Member	<i>2023</i>
FOCS Program Committee Member	<i>2022</i>

NeurIPS Area Chair	2019, 2021, 2022
AAAI Senior Area Chair	2022

Conference Reviewing.....

STOC, FOCS, SODA, ICALP, ITCS, ICML, NeurIPS, AAAI, AISTATS, SOCG, ESA, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS

Journal Reviewing.....

Journal of the ACM, SIAM Journal on Computing, ACM Transactions on Algorithms, Theoretical Computer Science, Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, Linear Algebra and Its Applications, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science, IEEE Journal on Selected Areas in Information Theory, BIT Numerical Mathematics, AMS Notices

Grants Panels.....

National Science Foundation Panel	2019, 2022
NSF Graduate Research Fellowship Reviewer	2022
Department of Energy Panel	2022

Notables.....

ICML 2018 Outstanding Reviewer, ICML Expert Review 2021

Invited Workshops and Events

Banff International Research Station (BIRS)

<i>Invited participant</i>	2023
Workshop on Perspectives on Matrix Computations: Theoretical Computer Science Meets Numerical Analysis.	

Rutgers, DIMACS

<i>Invited participant</i>	2019
Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.	

UC Berkeley, Simons Institute for the Theory of Computing

<i>Invited participant</i>	2018
Workshop on Randomized Numerical Linear Algebra and Applications.	

Japan National Institute of Informatics

<i>Invited participant</i>	2016
Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.	

Honors and Awards

NYU Tandon	2022
<i>Nominated for Jacobs Excellence in Education Innovation Award.</i>	

National Science Foundation	2014-2017
<i>Graduate Research Fellowship, recipient.</i>	

Yale University	2012
<i>Cum Laude, Distinction in Both Majors.</i>	

Other Professional Experience

Redfin	Seattle, WA
<i>Software Engineer, Data Team</i>	2012 – 2014

Elysium Digital
Summer Consultant

Amicus
Software Developer

Cambridge, MA
2011

New Haven, CT
2010 – 2011